

SECTION THREE

Creative Collaboration

CHAPTER 14

Set the Conditions for Team Creativity

by Greg Satell

One of the most damaging myths about creativity is that there is a specific creative personality that some people have and others don't. Yet in decades of creativity research, no such trait has ever been identified. The truth is that anybody can be creative, given the right opportunities and context.

If you don't believe me, take the least creative person in your office out for lunch—someone who doesn't seem to have a creative bone in their body. Chances are, you'll find some secret passion, pursued outside of office hours, into which they pour their creative energies. They just aren't applying those energies to their day jobs.

The secret to unlocking creativity is not to look for people who are more creative, but to unlock more creativity from the people who already work for you. The same body of creativity research that finds no distinct creative personality is incredibly consistent about what leads to creative work, and they are all things you can implement in your team.

Cultivating Expertise

Over the decades, creativity researchers have consistently found that expertise is absolutely essential for topnotch creative work—and the expertise needs to be specific to a particular field or domain. So the first step to being creative is to become an expert in a particular area.

To understand what a field's important problems are and what would constitute effective new solutions, you need to be an expert. Albert Einstein, for instance, studied physics intensely for years to understand the basic physical model for time and space before he understood that there was an inherent flaw in that model.

So how do you cultivate expertise? Performance expert Anders Ericsson studied that problem for decades and found that the crucial element is deliberate practice. You need to identify the components of a skill, offer coaching, and encourage employees to work on weak areas. This longer-term, continuous approach goes far beyond the intermittent training that most organizations do.

For example, one skill that Amazon has identified as crucial to performance is writing. Employees need to constantly write six-page memos, even for introducing small product features, throughout their careers at the company. They consistently receive coaching and feedback and need to write good memos to advance within the company.

Any company can replicate Amazon's memo-writing policy. What's not so easily replicated is the company's intense, long-term commitment to cultivating writing expertise.

Encouraging Exploration

While deep expertise in a given field is essential for real creativity, it is not sufficient. Look at any great body of creative work, and you'll find a crucial insight that came from outside the original domain. It is often a seemingly random piece of insight that transforms ordinary work into something very different. For example, a random visit to a museum inspired Pablo Picasso's African period. Charles Darwin spent years studying fossils and thinking about evolution until he came across a 40-year-old economics essay by Thomas Malthus. The essay led to Darwin's theory of natural selection. And the philosophy of David Hume helped lead Einstein to special relativity.

More recently, a team of researchers analyzing 17.9 million scientific papers found that the most highly cited work is far more likely to come from a team of experts in one field working with a specialist in something very different. That combination of expertise, exploration, and collaboration is what leads to truly breakthrough ideas.

Empowering Your People with Technology

In Walter Isaacson's biography of Leonardo da Vinci, he recounts how the medieval master would study nature, including anatomy and geological formations, to guide his art. Now, Leonardo was clearly a genius of historical proportions, but think about how much more efficient he would have been with a decent search engine.

One of the most overlooked aspects of innovation is the degree to which technology can enhance productivity. For one thing, technology greatly helps people acquire domain expertise and explore adjacencies. For another, it frees up time to allow for more experimentation.

You can see the power of technology at work at Pixar, which was originally a technology company that began shooting short films to demonstrate the capabilities of its original product, animation software. However, as the company was experimenting with the technology, it also found itself experimenting with storytelling, and those experiments led Pixar to become one of the most highly acclaimed studios in history.

As Pixar cofounder Ed Catmull put it in his memoir, *Creativity, Inc.*, "Every one of our films, when we start off, they suck. . . . Our job is to take it from something that sucks to something that doesn't suck. That's the hard part." This is the kind of continual iteration that technology facilitates—and that makes truly great creative work possible.

Rewarding Persistence

Far too often, we think of creativity as an initial brilliant spark followed by a straightforward period of execution, but as Catmull's preceding comment shows, that's not true in the least. In his book, he calls early ideas "ugly babies" and stresses the need to protect them from being judged too quickly. Yet most organizations do just the opposite. Any idea that doesn't show immediate promise is typically killed quickly and without remorse.

One firm that has been able to buck this trend is IBM. Its research division routinely pursues seemingly outlandish ideas long before they are commercially viable. For example, a team at IBM successfully performed the first quantum teleportation in 1993, when the company was in dire financial straits, with absolutely no financial benefit.

However, the research wasn't particularly expensive, and the company has continued to support the work for decades. Today, it is a leader in quantum computing—a market potentially worth billions—because it stuck with this technology. That's why IBM, despite its ups and downs, remains a highly profitable company while so many of its former rivals are long gone. (See [chapter 20](#) in this book for more on creative persistence.)

Kevin Ashton, who first came up with the idea for RFID chips, encourages persistence in his book *How to Fly a Horse*: "Creation is a long journey, where most turns are wrong and most ends are dead. The most important thing creators do is work. The most important thing they don't do is quit."

Yet all too often, organizations do quit. They expect their babies to be beautiful from the start. They see creation as an event rather than a process, fail to invest in expertise or exploration, and refuse to tolerate wrong turns and dead ends. Is it any wonder that so few companies produce anything truly new and different?

Greg Satell is an international keynote speaker, an adviser, and the bestselling author of *Cascades: How to Create a Movement That Drives Transformational Change* and *Mapping Innovation*. You can learn more about Greg on his website, GregSatell.com, and follow him on Twitter [@DigitalTonto](https://twitter.com/DigitalTonto).

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CHAPTER 15

Why Diverse Teams Are Smarter

by David Rock and Heidi Grant

In recent years, a body of research has revealed a nuanced benefit of workplace diversity: Nonhomogeneous teams are simply smarter. Working with people who are different from you may challenge your brain to overcome its stale ways of thinking and sharpen its performance and creativity. Let's dig into why diverse teams are smarter.

They Focus More on Facts

People from diverse backgrounds might actually alter the behavior of a group's social majority in ways that lead to improved and more accurate group thinking. In a study published in the *Journal of Personality and Social Psychology*, scientists assigned 200 people to six-person mock jury panels whose members were either all white or four white and two Black participants. The people were shown a video of a trial of a Black defendant and white victims. They then had to decide whether the defendant was guilty.

It turned out that, compared with the homogeneous panels, the diverse ones raised more facts related to the case and made fewer factual errors when discussing the available evidence. If errors did occur, they were more likely to be corrected during deliberation. One possible reason for this difference was that the white jurors on the diverse panels recalled evidence more accurately.

Other studies have yielded similar results. In a series of experiments conducted by Sheen S. Levine and colleagues in Texas and Singapore, scientists put financially literate people in simulated markets and asked them to price stocks. The participants were placed in either ethnically diverse or homogeneous teams. The researchers found that individuals who were part of the diverse teams were 58% more likely to price stocks correctly, whereas those in homogeneous groups were more prone to pricing errors.

Diverse teams are more likely to constantly reexamine facts and remain objective. They may also encourage greater scrutiny of each member's actions, keeping their joint cognitive resources sharp and vigilant. By breaking up workplace homogeneity, you can allow your employees to become more aware of their own potential biases—entrenched ways of thinking that can otherwise make them miss key information and even lead them to make errors in decision-making processes.

They Process Those Facts More Carefully

Greater diversity may also change the way that entire teams digest information needed to make the best decisions. In a study published in the *Personality and Social Psychology Bulletin*, Katherine Phillips of Northwestern University and her team divided groups of sorority or fraternity members into four-member groups, each of which had to read interviews conducted by a detective investigating a murder. Three people in every group, referred to as “old-timers” in the study, came from the same sorority or fraternity, whereas the fourth, the “newcomer,” was either a member of the same sorority or fraternity or a different one. The three old-timers in each group gathered to decide who was the most likely murder suspect. Five minutes into their discussion, the newcomer joined the deliberation and expressed an opinion as to who the suspect was.

Although groups with out-group newcomers felt less confident about the accuracy of their joint decisions, they were more likely to guess who the correct suspect was than were those with newcomers who belonged to the same group.

The scientists think that diverse teams may outperform homogeneous ones in decision-making because these teams process information more carefully. Remember: Considering the perspective of an outsider may seem counterintuitive, but the payoff can be huge.

They're Also More Innovative

To stay competitive, businesses should always continue to innovate. One of the best ways to boost their capacity to transform themselves and their products may involve hiring more women and culturally diverse teams, research suggests. In a study published in *Innovation: Management, Policy & Practice*, Cristina Díaz-García and colleagues analyzed levels of gender diversity in R&D teams from 4,277 companies in Spain. Using statistical models, they found that companies with more women were more likely to introduce radical new innovations into the market over a two-year period.

In another study, published in *Economic Geography*, Max Nathan and Neil Lee concluded that increased cultural diversity is a boon to innovativeness. They pooled data on 7,615 firms that participated in the London Annual Business Survey. This questionnaire asks London executives a number of questions about their companies' performance. The results revealed that businesses run by culturally diverse leadership teams were more likely to develop new products than were those with homogeneous leadership.

Though you may feel more at ease working with people who share your background, don't be fooled by your comfort. Hiring individuals who do not look, talk, or think like you can allow you to dodge the costly pitfalls of conformity, which discourages innovative thinking.

In a nutshell, enriching your employee pool with representatives of different genders, races, and nationalities is key for boosting your company's joint intellectual potential. Creating a more diverse workplace will help to keep your team members' biases in check and make them question their assumptions. At the same time, we need to make sure the organization has inclusive practices so that everyone feels they can be heard. All these efforts to encourage diversity can make your teams smarter and, ultimately, make your organization more successful, whatever your goals.

David Rock is cofounder of the NeuroLeadership Institute and author of *Your Brain at Work*. **Heidi Grant** is a social psychologist who researches, writes, and speaks about the science of motivation. Her most recent book is *Reinforcements: How to Get People to Help You*. She's also the author of *No One Understands You and What to Do About It* and *Nine Things Successful People Do Differently* (Harvard Business Review Press, 2018, 2015, 2012, respectively).

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CHAPTER 16

The Most Creative Teams Have a Specific Kind of Cultural Diversity

by Sujin Jang

Culturally diverse teams, research shows, can help deliver better outcomes in today's organizations. This is largely a good thing: Diverse teams have the potential to be more creative because of the breadth of information, ideas, and perspectives that members can bring to the table (see [chapter 15](#)). But because these teams often suffer from conflicting norms and differing assumptions between members, they don't always reach their full creative potential. When managers don't know how to spot and address these situations, cultural diversity may actually inhibit a team's creative performance.

The Importance of Cultural Brokers

My research, which was published in *Organization Science*, finds that *cultural brokerage* is a key factor that allows multicultural teams to capitalize on the benefits of diversity while mitigating the pitfalls. I define cultural brokerage as the act of facilitating interactions across parties from different cultural backgrounds. In two studies—an archival study of over 2,000 multicultural teams and an experiment involving 83 multicultural teams with different cultural compositions—I found that teams were significantly more creative when they had one or more members who acted as a cultural broker.

Who are these cultural brokers? They're team members who have relatively more multicultural experience than others do and who act as a bridge between their monocultural teammates. These brokers come in two profiles. First, they can have multicultural experiences that map directly onto the cultures they are bridging. For example, in a team with mostly Indian and American team members, a cultural broker could be someone with experience in both Indian and American cultures. I call such individuals *cultural insiders*. The second type of cultural broker is someone with experience in two or more cultures not represented in the team—say, Australian and Korean. I call such individuals *cultural outsiders*.

I found that cultural insiders and outsiders each draw on their distinct cultural backgrounds relative to the team to engage in different kinds of cultural brokerage. In the experimental study, cultural insiders used their dual knowledge of the other cultures on their team to integrate information and ideas from those cultures. In other words, they often proposed ideas that combined elements of both cultures. Meanwhile, cultural outsiders drew on their position as a neutral third party to elicit information and ideas from the other cultures represented in the team. They tended to ask questions of other team members and invite them to share relevant cultural knowledge. Both types of cultural brokerage led to a boost in creativity at the team level.

Why Cultural Diversity Alone Is Not Enough

First and foremost, this research suggests that it is not enough to simply bring together people from different cultures and expect them to produce creative outcomes. For teams to unleash their full creative potential, they need at least one multicultural insider or outsider in the group. Outsider cultural brokers, I suspect, may be less common in

organizations, because many people incorrectly assume only those with culture-specific knowledge are in a position to facilitate cross-cultural interactions. In fact, however, cultural outsiders are just as effective as cultural insiders are in enhancing team creativity. This is particularly good news for highly diverse teams, which are unlikely to have a single cultural insider.

At the same time, you cannot simply assign someone to be a cultural broker and call it day. A formal appointment does not guarantee that a person will be effective; instead, organizations should take care to create the conditions that can allow cultural brokerage to emerge. Remember, being a broker requires substantial cognitive and emotional effort. Because of these requirements, effective cultural brokerage is more likely to emerge in teams with a high level of psychological safety. It also requires active participation and buy-in from the team as a whole and is more likely to emerge in teams that view diversity as a resource and a source of learning.

Collaboration in multicultural teams is a complex and multifaceted endeavor. While this type of work can be challenging, my research suggests that understanding the dynamics of cultural brokerage provides a critical advantage in realizing the creative potential of diverse teams.

Sujin Jang is an assistant professor of organisational behaviour at INSEAD. Her research focuses on global teams and the challenges of working across cultures.

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CHAPTER 17

How to Cultivate Psychological Safety on Your Team

by Laura Delizonna

“There’s no team without trust,” says Paul Santagata, head of industry at Google. He knows the results of the tech giant’s massive two-year study on team performance, which revealed that the highest-performing teams have one thing in common: psychological safety, the belief that you won’t be punished when you make a mistake. Studies show that psychological safety allows for moderate risk-taking, speaking your mind, creativity, and sticking your neck out without fear of having it cut off—just the types of behavior that lead to market breakthroughs.

Ancient evolutionary adaptations explain why psychological safety is both fragile and vital to success in uncertain, interdependent environments. The brain processes a provocation by a boss, a competitive coworker, or a dismissive subordinate as a life-and-death threat. The amygdala, the alarm bell in the brain, ignites the fight-or-flight response, hijacking higher brain centers. This “act first, think later” brain structure shuts down perspective and analytical reasoning. Quite literally, just when we need it most, we lose our minds. While that fight-or-flight reaction may save us in life-and-death situations, it handicaps the strategic thinking needed in today’s workplace.

Twenty-first-century success depends on another system—the broaden-and-build mode of positive emotion, which allows us to solve complex problems and foster cooperative relationships. Barbara Fredrickson at the University of North Carolina has found that positive emotions like trust, curiosity, confidence, and inspiration broaden the mind and help us build psychological, social, and physical resources. We become more open-minded, resilient, motivated, and persistent when we feel safe. Humor increases, as does our ability to find solutions and apply divergent thinking—the cognitive process underlying creativity.

When the workplace feels challenging but not threatening, teams can sustain the broaden-and-build mode. Oxytocin levels in our brains rise, eliciting trust and trust-making behavior. Trust is a huge factor in team success, as Santagata attests: “In Google’s fast-paced, highly demanding environment, our success hinges on the ability to take risks and be vulnerable in front of peers.”

So how can you increase psychological safety on your own team? Try replicating the steps that Santagata took with his.

We humans hate losing even more than we love winning. A perceived loss triggers attempts to reestablish fairness through competition, criticism, or disengagement, which is a form of workplace-learned helplessness. Santagata knows that true success is a win-win outcome, so when conflicts come up, he avoids triggering a fight-or-flight reaction by asking, “How could we achieve a mutually desirable outcome?”

Underlying every team’s who-did-what confrontation are universal needs such as respect, competence, social status, and autonomy. Recognizing these deeper needs naturally elicits trust and promotes positive language and behaviors. Santagata reminded his team that even in the most contentious negotiations, the other party is just like them and aims to walk away happy. He led them through a reflection called Just Like Me, which asks you to consider these likely observations:

- This person has beliefs, perspectives, and opinions, just like me.
- This person has hopes, anxieties, and vulnerabilities, just like me.

- This person has friends, family, and perhaps children who love them, just like me.
- This person wants to feel respected, appreciated, and competent, just like me.
- This person wishes for peace, joy, and happiness, just like me.

“Thinking through in advance how your audience will react to your messaging helps ensure your content will be heard, versus your audience hearing an attack on their identity or ego,” explains Santagata.

Skillfully confront difficult conversations head-on by preparing for likely reactions. For example, you may need to gather concrete evidence to counter defensiveness when discussing hot-button issues. Santagata asks himself, “If I position my point in this manner, what are the possible objections, and how would I respond to those counterarguments?” He says, “Looking at the discussion from this third-party perspective exposes weaknesses in my positions and encourages me to rethink my argument.” Specifically, he asks three questions:

- What are my main points?
- What are three ways my listeners are likely to respond?
- How will I respond to each of those scenarios?

If team members sense that you’re trying to blame them for something, you become their saber-toothed tiger. John Gottman’s research at the University of Washington shows that blame and criticism reliably escalate conflict, leading to defensiveness and—eventually—to disengagement. The alternative to blame is curiosity. If you believe you already know what the other person is thinking, then you’re not ready to have a conversation. Instead, adopt a learning mindset, knowing you don’t have all the facts. Here’s how:

- State the problematic behavior or outcome as an observation, and use factual, neutral language. For example, “In the past two months, there’s been a noticeable drop in your participation during meetings, and progress appears to be slowing on your project.”
- Engage them in an exploration. For example, “I imagine there are multiple factors at play. Perhaps we could uncover what they are together?”
- Ask for solutions. The people who are responsible for creating a problem often hold the keys to solving it. That’s why a positive outcome typically depends on their input and buy-in. Ask directly, “What do you think needs to happen here?” Or, “What would be your ideal scenario?” Another question leading to solutions is, “How could I support you?”

Asking for feedback on how you delivered your message disarms your opponent, illuminates blind spots in communication skills, and models fallibility. All of these results increase trust in leaders. Santagata closes difficult conversations with these questions:

- What worked and what didn’t work in my delivery?
- How did it feel to hear this message?
- How could I have presented it more effectively?

For example, Santagata asked about his delivery after giving his senior manager tough feedback. His manager replied, “This could have felt like a punch in the stomach, but you presented reasonable evidence, and that made me want to hear more. You were also eager to discuss the challenges I had, which led to solutions.”

Santagata periodically asks his team how safe they feel and what could enhance their feeling of safety. In addition, his team routinely takes surveys on psychological safety and other team dynamics. Some teams at Google include questions such as, “How confident are you that you won’t receive retaliation or criticism if you admit an error or make a mistake?”

If you create this sense of psychological safety on your own team starting now, you can expect to see higher levels of engagement. Your group will also enjoy increased motivation for tackling difficult problems, more learning and development opportunities, and better performance.

Laura Delizonna is an executive coach, an instructor at Stanford University, a keynote speaker, and a culture consultant at Delizonna.com. She specializes in equipping leaders of top companies with the frameworks and tools to build high-performance cultures.

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CHAPTER 18

How to Give and Receive Feedback About Creative Work

by Spencer Harrison

Feedback is crucial for learning and improving, but we rarely enjoy being on the receiving end of it when it's criticism. Many people have a negative reaction to feedback, especially feedback on their creative work. In a study of seven companies and 11,471 days of creative work, Harvard Business School professor Teresa M. Amabile and her colleagues found two striking patterns: First, getting feedback was incredibly rare, indicating that people seemed to avoid it. And second, when people did receive feedback, it generally left a negative emotional residue.

So, what might good feedback for creative work look like? By *good feedback*, I mean feedback that people actually want and that leads to changes that improve their creative output.

Identifying how to do this requires understanding how creativity works. Creativity is the generation of an idea that is both useful and novel. Combining the two requires some care because novelty, by definition, is something unfamiliar to both the creator and anyone seeing the idea for the first time. As a result, early creative ideas can be fragile and dismissed as too new, weird, or unnecessary. New ideas need direction that can build them up, rather than critiques that can tear them down.

At the same time, creativity does require feedback. As much as we have mythologized creativity as the domain of an individual genius working alone, almost all creative processes used in organizations—design thinking, lean startup methodologies, agile development, and more—benefit from feedback early on. As a result, organizations must provide effective feedback to cultivate creative ideas; it's one of the ways for them to adapt to industry changes and competitive pressures. Leaders will want to understand how to give and receive feedback effectively in creative work.

Karyn Dossinger and I have published research that gets at this question, focusing on a successful online company that crowdsources T-shirt designs from a large community of freelance designers. Its website hosts a forum where designers can ask for feedback on early design prototypes from their peers and then post updated designs based on the comments they received. Looking across almost 2,000 feedback statements, we learned that two dynamics seem crucial: First, designers who were motivated to seek feedback out of curiosity, as opposed to simply improving their design, attracted more and higher-quality feedback. Second, the peer critics who recognized that feedback is a subjective opinion as opposed to an objective statement were more effective in enhancing the creativity of the final design. Let's examine these two dynamics.

Asking for Feedback Out of Curiosity

How we ask for people's opinions influences the scope and type of feedback we receive. Sometimes the requests are overly narrow. For example, we found that some T-shirt designers asked about the color of a dinosaur they had illustrated, their font choice, or the placement of a monkey working on a typewriter. There are often underlying reasons for asking a specific question like this. You might, for example, be trying to restrain a coworker from attacking your work, or perhaps you're showcasing something you're proud of (in which case you really don't want feedback—you want admiration).

This narrow approach, however, limits the potential of creative work, because it doesn't allow for the possibility of novelty. Changing one color, for example, may not push the boundaries to create something that peers and potential customers haven't seen before.

Our research showed that highly curious individuals asked extremely open questions like "What do you think?" or "Where could I go next with this?" These designers received significantly more feedback than those asking narrow questions, and their final designs received higher scores. By leading with curiosity, the feedback seekers signaled an openness to ideas beyond their own and enjoyed the benefits of broader ideas. In contrast, narrow questions signal that the feedback seeker already has a set of ideas and wants validation that those ideas are correct. In this way, creative work is like dancing: Questions born out of curiosity signal that the creative worker is looking for a dance partner. (For more on the business case for curiosity, see [chapter 21](#) of this guide.)

Providing Feedback Based on Subjectivity

If asking questions is like asking for a dance partner, then providing feedback is being the type of dance partner someone would actually want to dance with. A huge part of creativity is this ability to get your audience to agree that what you're doing feels new and surprising to them. And just as asking open questions honors the fact that creativity requires novelty, providing feedback needs to honor that same assumption. It needs to provide space for something new, something that might not be anticipated by either party, to emerge.

When providing feedback to creative workers, signal that your opinion is exactly that: an opinion. This acknowledgment of your subjectivity seems deceptively easy. It requires providing feedback that includes first-person pronouns: *I*, *me*, and *my*. "I see . . ." or "What strikes me is that . . ." or "My opinion is . . ." Many managers find this subjective angle difficult, because they have been trained to solve concrete problems, not to consider what something really means. Providing feedback on creative work means setting aside the managerial impulse to plan and retain control. By speaking from a personal point of view, managers recognize that their opinions provide *potential trajectories* a creative worker might try—not the "right" road to take.

Feedback is tricky, and it's even trickier for creative work. But asking for feedback out of curiosity and basing your feedback on subjectivity can improve both the process and the outcomes.

Spencer Harrison is an associate professor of organisational behaviour at INSEAD. He grew up drawing cartoons, invents stories for his kids, likes using the word *puzzle* as a verb, and researches creativity and how people connect to their work.

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CHAPTER 19

Why Design Thinking Works

by Jeanne Liedtka

Occasionally, a new way of organizing work leads to extraordinary improvements. Total quality management did that in manufacturing in the 1980s by combining a set of tools—kanban cards, quality circles, and so on—with the insight that people on the shop floor could do much higher level work than they were usually asked to. That blend of tools and insight, applied to a work process, can be thought of as a *social technology*.

In a seven-year study, I looked in depth at 50 projects from a range of sectors, including business, health care, and social services. I found that another social technology, design thinking, has the potential to do for innovation exactly what total quality management did for manufacturing: unleash people's full creative energies, win their commitment, and radically improve processes. By now most executives have at least heard about design thinking's tools—ethnographic research, an emphasis on reframing problems and experimentation, the use of diverse teams, and so on—if not tried them. But what people may not understand is the subtler way that design thinking gets around the human biases (for example, rootedness in the status quo) or attachments to specific behavioral norms (“That’s how we do things here”) that time and again block the exercise of imagination.

In this chapter, I’ll explore a variety of human tendencies that get in the way of innovation, and I’ll describe how design thinking’s tools and clear process steps help teams break free of them. Let’s begin by looking at what organizations need from innovation—and at why their efforts to obtain it often fall short.

The Challenges of Innovation

To be successful, an innovation process must deliver three things: superior solutions, lower risks and costs of change, and employee buy-in. Over the years, businesspeople have developed useful tactics for achieving those outcomes. But when trying to apply them, organizations frequently encounter new obstacles and trade-offs.

Superior solutions

Defining problems in obvious, conventional ways, not surprisingly, often leads to obvious, conventional solutions. *Asking a more interesting question* can help teams discover more-original ideas. The risk is that some teams may get indefinitely hung up exploring a problem, while action-oriented managers may be too impatient to take the time to figure out what question they should be asking.

It’s also widely accepted that solutions are much better when they incorporate *user-driven criteria*. Market research can help companies understand those criteria, but the hurdle here is that it’s hard for customers to know they want something that doesn’t yet exist.

Finally, bringing *diverse voices* into the process is also known to improve solutions. This can be difficult to manage, however, if conversations among people with opposing views deteriorate into divisive debates.

Lower risks and costs

Uncertainty is unavoidable in innovation. That's why innovators often build a *portfolio of options*. The trade-off is that a portfolio of too many ideas dilutes focus and resources. To manage this tension, innovators must be willing to let go of bad ideas—to “call the baby ugly,” as a manager in one of my studies described it. Unfortunately, people often find it easier to kill the creative (and arguably riskier) ideas than to kill the incremental ones.

Employee buy-in

An innovation won't succeed unless a company's employees get behind it. The surest route to winning their support is to involve them in the generation of ideas. The danger is that the involvement of many people with different perspectives will create chaos and incoherence.

Underlying the trade-offs associated with achieving these outcomes is a more fundamental tension. In a stable environment, organizations achieve efficiency by driving variation out of their processes. But in an unstable world, variation becomes the organization's friend because it opens new paths to success. However, who can blame leaders who must meet quarterly targets for doubling down on efficiency, rationality, and centralized control?

To manage all the trade-offs, organizations need a social technology that addresses these behavioral obstacles and the counterproductive biases of human beings. And as I'll explain next, design thinking fits that bill.

The Beauty of Structure

Experienced designers often complain that design thinking is too structured and linear. And for them, that's certainly true. But managers on innovation teams are generally not designers. They're not used to doing face-to-face research with customers, getting deeply immersed in their perspectives, cocreating with stakeholders, and designing and executing experiments. Structure and linearity help managers try to adjust to these new behaviors.

As Kaaren Hanson, formerly the head of design innovation at Intuit and now chief design officer at JPMorgan Chase, explains, “Anytime you're trying to change people's behavior, you need to start them off with a lot of structure, so they don't have to think. A lot of what we do is habit, and it's hard to change those habits, but having very clear guardrails can help us.”

Organized processes keep people on track and curb the tendency to spend too long exploring a problem or to impatiently skip ahead. They also instill confidence. Most humans are driven by a fear of mistakes, so they focus more on preventing errors than on seizing opportunities. When a choice risks failure, they opt for inaction rather than action. But because there is no innovation without action, psychological safety is essential. The physical props and highly formatted tools of design thinking deliver that sense of security, helping would-be innovators move more assuredly through the discovery of customer needs, idea generation, and idea testing.

In most organizations the application of design thinking involves seven activities. Each generates a clear output that the next activity converts to another output until the organization arrives at an implementable innovation. But at a deeper level, something else is happening—something that executives are generally unaware of. Though ostensibly geared to understanding and molding the experiences of customers, each design-thinking activity also profoundly reshapes the experiences of the *innovators themselves*.

Customer Discovery

Many of the best-known methods of the design-thinking discovery process relate to identifying the “job to be done.” Adapted from the fields of ethnography and sociology, these methods concentrate on examining what makes for a meaningful customer journey rather than focusing on the collection and analysis of data. This exploration entails three sets of activities.

Immersion

Traditionally, customer research has been an impersonal exercise. An expert, who may well have preexisting theories about customer preferences, reviews feedback from focus groups, surveys, and, if available, data on current behavior and draws inferences about customer needs. The better the data, the better the inferences. The trouble is, this approach grounds people in the already-articulated needs that the data reflects. They see the data through the lens of their own biases. And they don’t recognize needs that people have *not* expressed.

Design thinking takes a different approach: The innovator identifies hidden needs by living the customer’s experience. Consider what happened at the Kingwood Trust, a U.K. charity helping adults with autism and Asperger’s syndrome. One design team member, Katie Gaudion, got to know Pete, a nonverbal adult with autism. The first time she observed him at his home, she saw him engaged in seemingly damaging acts—like picking at a leather sofa and rubbing indents in a wall. She started by documenting Pete’s behavior and defined the problem as how to prevent such destructiveness.

But on her second visit to Pete’s home, she asked herself, “What if Pete’s actions were motivated by something other than a destructive impulse?” Putting her personal perspective aside, she mirrored his behavior and discovered how satisfying his activities actually felt. “Instead of a ruined sofa, I now perceived Pete’s sofa as an object wrapped in fabric that is fun to pick,” she explained. “Pressing my ear against the wall and feeling the vibrations of the music above, I felt a slight tickle in my ear whilst rubbing the smooth and beautiful indentation. . . . So instead of a damaged wall, I perceived it as a pleasant and relaxing audio-tactile experience.”

Gaudion’s immersion in Pete’s world not only produced a deeper understanding of his challenges but also called into question an unexamined bias about the residents, who had been perceived as disability sufferers who needed to be kept safe. Her experience caused her to ask herself another new question: Instead of designing just for residents’ disabilities and safety, how could the innovation team design for their strengths and pleasures? That led to the creation of living spaces, gardens, and new activities aimed at enabling people with autism to live fuller and more pleasurable lives. [Table 19-1](#) summarizes how innovators take on this immersive approach.

Sense making

Immersion in user experiences provides raw material for deeper insights. But finding patterns and making sense of the mass of qualitative data collected is a daunting challenge. Time and again, I have seen initial enthusiasm about the results of ethnographic tools fade as nondesigners become overwhelmed by the volume of information and the messiness of searching for deeper insights. It is here that the structure of design thinking really comes into its own.

TABLE 19-1

Shaping the innovator’s journey

What makes design thinking a social technology is its ability to counteract the biases of innovators and change the way they engage in the innovation process.

Problem	Design thinking	Improved outcome
Innovators are trapped in their own expertise and experience.	Design thinking provides immersion in the user's experience, shifting an innovator's mindset toward . . .	. . . a better understanding of those being designed for.
Innovators feel overwhelmed by the volume and messiness of qualitative data.	Design thinking makes sense of data by organizing it into themes and patterns, pointing the innovator toward . . .	. . . new insights and possibilities.
Innovators are divided by differences in team members' perspectives.	Design thinking builds alignment as insights are translated into design criteria, moving an innovation team toward . . .	. . . convergence around what really matters to users.
Innovators are confronted by too many disparate but familiar ideas.	Design thinking encourages the emergence of fresh ideas through a focused inquiry, shifting team members toward . . .	. . . a limited but diverse set of potential new solutions.
Innovators are constrained by existing biases about what does or doesn't work.	Design thinking fosters articulation of the conditions necessary to each idea's success and transitions a team toward . . .	. . . clarity on make-or-break assumptions; enables the design of meaningful experiments.
Innovators lack a shared understanding of new ideas and are often unable to get good feedback from users.	Design thinking offers pre-experiences to users through rough prototypes that help innovators get . . .	. . . accurate feedback at low cost and an understanding of potential solutions' true value.
Innovators fear change and ambiguity surrounding the new future.	Design thinking delivers learning in action as experiments engage staff and users, helping them build . . .	. . . a shared commitment and confidence in the new product or strategy.

One of the most effective ways to make sense of the knowledge generated by immersion is a design-thinking exercise called the gallery walk. In this exercise, the core innovation team selects the most important data gathered during the discovery process and writes it down on large posters. Often these posters showcase individuals who have been interviewed, complete with their photos and quotations capturing their perspectives. The posters are hung around a room, and key stakeholders are invited to tour this gallery and write down on Post-it notes the bits of data they consider essential to new designs. The stakeholders then form small teams, and in a carefully orchestrated process, their Post-it observations are shared, combined, and sorted by theme into clusters that the group mines for insights. This process overcomes the danger that innovators will be unduly influenced by their own biases and see only what they want to see, because it makes the people who were interviewed feel vivid and real to those browsing the gallery. The exercise creates a common database and facilitates collaborators' ability to interact, reach shared insights together, and challenge one another's individual takeaways—another critical guard against biased interpretations.

Alignment

The final stage in the discovery process is a series of workshops and seminar discussions that ask some form of this question: If anything were possible, what job would the design do well? The focus on possibilities rather than on the constraints imposed by the status quo helps diverse teams have more-collaborative and creative discussions about the design criteria, that is, the key features that an ideal innovation should have. Establishing a spirit of inquiry deepens dissatisfaction with the status quo and makes it easier for teams to reach consensus throughout the innovation process. And down the road, when the portfolio of ideas is winnowed, agreement on the design criteria will give novel ideas a fighting chance against safer, incremental ones.

Consider what happened at Monash Health, an integrated hospital and health-care system in Melbourne, Australia. Mental health clinicians there had long been concerned about the frequency of patient relapses—usually in the form of drug overdoses and suicide attempts—but consensus on how to address this problem eluded them. In an effort to get to the bottom of it, clinicians traced the experiences of specific patients through their treatment. One patient, Tom, emerged as emblematic in their study. His experience included three face-to-face visits with different clinicians, 70 touchpoints, 13 different case managers, and 18 handoffs during the interval between his initial visit and his relapse.

The team members held a series of workshops in which they asked clinicians this question: Did Tom's current care exemplify why they had entered health care? As people discussed their motivations for becoming doctors and nurses, they came to realize that improving Tom's outcome might depend as much on their sense of duty to Tom himself as it did on their clinical activity. Everyone bought into this conclusion, which made designing a new treatment process—centered on the patient's needs rather than perceived best practices—proceed smoothly and successfully. After its implementation, patient-relapse rates fell by 60%.

Idea Generation

Once they understand customers' needs, innovators move on to identify and winnow down specific solutions that conform to the criteria they've identified.

Emergence

The first step here is to set up a dialogue about potential solutions, carefully planning who will participate, what challenge they will be given, and how the conversation will be structured. After using the design criteria to do some individual brainstorming, participants gather to share ideas and build on them creatively—as opposed to simply negotiating compromises when differences arise.

When Children’s Health System of Texas, the sixth-largest pediatric medical center in the United States, identified the need for a new strategy, the organization, led by the vice president of population health, Peter Roberts, applied design thinking to reimagine its business model. During the discovery process, clinicians set aside their bias that what mattered most was medical intervention. They came to understand that intervention alone wouldn’t work if the local population in Dallas lacked the time or ability to seek out medical knowledge and had no strong support networks—something few families in the area enjoyed. The clinicians also realized that the medical center couldn’t successfully address problems on its own; the community would need to be central to any solution. So Children’s Health invited its community partners to codesign a new wellness ecosystem whose boundaries (and resources) would stretch far beyond the medical center. Deciding to start small and tackle a single condition, the team gathered to create a new model for managing asthma.

The session brought together hospital administrators, physicians, nurses, social workers, parents of patients, and staff from Dallas’s school districts, its housing authority, the YMCA, and faith-based organizations around the city. First, the core innovation team shared learning from the discovery process. Next, each attendee thought independently about the capabilities that their own institution might contribute toward addressing the children’s problems, jotting down ideas on sticky notes. The participants were then invited to join a small group at one of five tables, where people would share individual ideas, group them into common themes, and envision what an ideal experience would look like for the young patients and their families.

Champions of change usually emerge from these kinds of conversations, and their presence greatly improves the chances of successful implementation. (All too often, good ideas die on the vine in the absence of people with a personal commitment to making them happen.) At Children’s Health, the partners invited into the project galvanized the community to act. In their institutions, they also forged and maintained the relationships required to realize the new vision. Housing authority representatives drove changes in housing codes, charging inspectors with incorporating children’s health issues (like the presence of mold) into their assessments. Local pediatricians adopted a set of standard asthma protocols, and parents of children with asthma took on a significant role as peer counselors providing intensive education to other families through home visits.

Articulation

Typically, emergence activities generate several competing ideas, more or less attractive and more or less feasible. In the next step, articulation, innovators uncover and examine their implicit assumptions. Managers are often bad at articulation, because of many behavioral biases, such as overoptimism, confirmation bias, and fixation on first solutions. When assumptions aren’t challenged, discussions around what will or won’t work become deadlocked, with everyone advocating from their own unique understanding of how the world works.

In contrast, design thinking frames the discussion as an inquiry into what would have to be true about the world for an idea to be feasible. An example of this approach comes from the Ignite Accelerator program of the U.S. Department of Health and Human Services. At the Whiteriver Indian Hospital in Arizona, a team led by Marliza Rivera, a young quality-control officer, sought to reduce the hospital's emergency room wait times, which were sometimes as long as six hours.

The team's initial concept, borrowed from the Johns Hopkins Hospital in Baltimore, was to install an electronic kiosk for check-in. As team members began to apply design thinking, however, they were asked to surface their assumptions about why the idea would work. It was only then that they realized that their patients, many of whom were older Apache speakers, were unlikely to be comfortable with computer technology. Approaches that worked in urban Baltimore would not work in Whiteriver, so this idea could be safely set aside.

At the end of the idea generation process, innovators will have a portfolio of well-thought-out but possibly quite different ideas. The assumptions underlying them will have been carefully vetted, and the conditions necessary for their success will be achievable. The ideas will also have the support of committed teams, who will be prepared to take on the responsibility of bringing them to market.

The Testing Experience

Companies often regard prototyping as a process of fine-tuning a product or service that has already largely been developed. But in design thinking, prototyping is carried out on far-from-finished products. It's about users' iterative experiences with a work in progress. The project's malleable status means that quite radical changes—including complete redesigns—can occur along the way.

Pre-experience

Neuroscience research indicates that helping people *pre-experience* something novel—that is, imagine it incredibly vividly—results in more-accurate assessments of the novelty's value. That's why design thinking calls for the creation of basic, low-cost artifacts that will capture the essential features of the proposed user experience. These are not literal prototypes—and they are often much rougher than the minimum viable products that lean startups test with customers. But what these artifacts lose in fidelity they gain in flexibility, because they can easily be altered in response to what the users learned when exposed to them. And their incompleteness invites interaction.

Such artifacts can take many forms. Designers tested the layout of a new medical office building at Kaiser Permanente, for example, by hanging bedsheets from the ceiling to mark future walls. Nurses and physicians were invited to interact with staffers who were playing the role of patients and to suggest how the spaces could be adjusted to better facilitate treatment. At Monash Health, a program called Monash Watch was aimed at using telemedicine to keep vulnerable populations healthy at home and reduce their hospitalization rates. The designers of the program used detailed storyboards to help hospital administrators and government policy makers envision this new approach in practice, without building a digital prototype.

Learning in action

Real-world experiments are an essential way to assess new ideas and identify the changes needed to make them workable. But such tests offer another, less obvious kind of value: They help reduce employees' and customers' quite normal fear of change.

Consider an idea proposed by Donald Campbell, a professor of medicine, and Keith Stockman, then a manager of operations research at Monash Health. As part of Monash Watch, they suggested hiring laypeople to be telecare guides who would act as “professional neighbors,” maintaining frequent telephone contact with patients at high risk of multiple hospital admissions. Campbell and Stockman hypothesized that lower-wage laypeople who were carefully selected, trained in health literacy and empathy skills, and backed by a decision support system and professional coaches the lay guides could involve as needed could help keep the at-risk patients healthy at home.

Their proposal was met with skepticism. Many of their colleagues held a strong bias against letting anyone besides a health professional perform such a service for patients with complex issues, but using health professionals in the role would have been unaffordable. Rather than debating this point, however, the innovation team members acknowledged the concerns and engaged their colleagues in the codesign of an experiment testing that assumption. Three hundred patients later, the results were in: Overwhelmingly positive patient feedback and a demonstrated reduction in bed use and emergency room visits, corroborated by independent consultants, quelled the fears of the skeptics.

As we have seen, the structure of design thinking creates a natural flow from research to rollout. Immersion in the customer experience produces data, which is transformed into insights that help teams agree on the design criteria they use to brainstorm solutions. Assumptions about what’s critical to the success of those solutions are examined and then tested with rough prototypes that help teams further develop innovations and prepare them for real-world experiments.

Along the way, design-thinking processes counteract human biases that thwart creativity while addressing the challenges typically faced in reaching superior solutions, lowered costs and risks, and employee buy-in. Recognizing organizations as collections of human beings who are motivated by varying perspectives and emotions, design thinking emphasizes engagement, dialogue, and learning. By involving customers and other stakeholders in the definition of the problem and the development of solutions, design thinking garners a broad commitment to change. And by supplying a structure to the innovation process, design thinking helps innovators collaborate and agree on what is essential to the outcome at every phase. It does this not only by overcoming workplace politics but also by shaping the experiences of the innovators, the key stakeholders, and the key implementers at every step. *That* is social technology at work.

Jeanne Liedtka is a professor of business administration at the University of Virginia’s Darden School of Business.

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CHAPTER 20

Your Best Ideas Are Often Your Last Ideas

by Loran Nordgren and Brian Lucas

To live up to our creative potential, we must first understand how the creative process actually works. Prior research has shown that people's first ideas are rarely their most creative. Coming up with just one breakthrough idea typically requires a lengthy brainstorming process in which you generate a large pool of potential options and iterate on them before finally reaching your most creative idea.

Despite this reality, however, most people consistently underestimate the value of persistence in the creative process. In our research, we document a fallacy we call the *creative cliff illusion*: Although creativity in fact tends to either increase or stay the same across an ideation session, people assume that creativity drops off over time.

We explored this misconception—and what managers can do to overcome it—with a series of studies in which we asked people to predict how their creativity would change over an ideation session. We ran these experiments using a variety of creative problem-solving tasks and across several sample U.S. populations, including university students and working adults.

In one of our studies, we enrolled professional and amateur comedians in a caption-writing competition. We surveyed their beliefs about how creativity changes over time and then had them spend as long as they wanted coming up with captions for a cartoon. We found that the comedians who were more certain that their early ideas would be their best ones stopped ideating sooner. These comedians ended up submitting fewer jokes. And importantly, fewer of their submitted jokes were rated as being highly creative. These results suggest that if you think your first ideas will be your best, you're more likely to stop the creative process before your actual best ideas are uncovered.

Why are people so bad at predicting their own creativity? The answer lies with a performance indicator that is often (falsely) conflated with creativity: productivity. When we are generating ideas (or, for that matter, when we are completing any task that requires mental energy), productivity does tend to decline over time, and people often take the ease of producing ideas as a signal of the quality of those ideas. Because people recognize that their productivity declines with time, they think the creativity of the ideas they produce must decline as well. But of course, this is not the case. In the creative process, the ideas that are most easily accessed tend to be the most obvious ones. Only by digging deeper do we finally uncover more novel, creative ideas.

GIVING UP IS THE ENEMY OF CREATIVITY

1. Ignore your first instinct to stop. When working on a tough creative challenge, you will be likely to face a moment when you feel stuck and can't come up with any more ideas. You'll first want to quit and spend your time doing something else. Temporarily ignore this instinct, especially if you're still in the early stages of the work. Try to generate just a few more ideas, or consider just a few more alternatives. You may find that your next creative idea was closer than you imagined.

2. Remember that creative problems are supposed to feel difficult. Most involve setbacks, failures, and that stuck feeling. These feelings are part of the process. Suppress your instinct to interpret them as a signal that you just aren't creative or that you've run out of good ideas. Reaching your creative potential will often take time, you'll need persistence to see a challenge through to the end.

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Luckily, not everyone is equally susceptible to the creative cliff illusion. In another study from the same series of experiments, we asked participants to report how frequently they engaged in creative work in their daily lives. We found that people who reported doing this kind of work more frequently were less likely to assume that creativity declined with time. Our finding suggests that personal experience with creative tasks can help people overcome their flawed assumptions about the creative process. In a third study, we found that regardless of past experience, simply telling people about this illusion can help to attenuate it.

Given these findings, there are a few things that managers can do to help their teams reset their expectations about the creative process and empower people to develop more creative ideas.

Educating Your People

Our research showed how pervasive the creative cliff illusion is—but it also illustrated that a better understanding of the phenomenon can reduce its potency. Managers should educate their teams about this illusion, explaining that a person's first ideas are unlikely to be as valuable as those that emerge later in the process. A manager can have one-on-one conversations with key employees or conduct teamwide discussions about how project workflows could be adapted to overcome this first-ideas bias.

In addition, managers can remind people that when their productivity declines and they begin to feel as if they're running out of ideas, this feeling doesn't mean that they are running low on creativity. While this sense of ineffectuality may be uncomfortable, it is actually the mind pushing its cognitive boundaries and searching for novel connections—two key conditions for generating creative ideas.

Investing in Your Creative Process

There is no set formula for success (such as “committing X% more time to the ideation process will give you Y% more creative ideas”). But there are several ways to ensure you're giving key creative processes the time and attention they need:

- Set aside more time for creative processes than you might think is necessary—whether that means an extra ideation session, a longer brainstorm, or even dedicated buffer time that can be used for additional meetings if needed.
- Ask your team to generate two or even three times as many ideas as you think you need, particularly in the early stages of ideation. Setting aggressive idea quotas can help people push past more obvious, early-stage ideas and uncover truly novel opportunities.

- Experiment with your team's creative processes, and measure the results. You can then use that data to drive future decisions. For example, the next time you run a workshop, track when the best ideas were generated. Did they come from the team that brainstormed for one hour or the team that took three hours? Did increasing your team's idea quota produce more highly creative proposals? Testing these different variables can help you calibrate your process and capture your team's creative potential.

The creative cliff illusion is very common—but it's also something that managers can help their teams overcome through a combination of education and process improvements. With the strategies outlined here, leaders can help their people understand how creativity works and build a process that emphasizes the value of late-stage ideas. Although it may take some extra work, the effort will be rewarded when you reach the potentially game-changing ideas that are uncovered through a lengthier ideation process.

Loran Nordgren is a professor of management and organizations at the Kellogg School of Management. **Brian Lucas** is an assistant professor at the Industrial and Labor Relations School, Cornell University.

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